16 Using algebra, prove that $0.1\dot{3}\dot{6} \times 0.\dot{2}$ is equal in value to $\frac{1}{33}$

Let $x = 0.1363636...$

$1000x = 136.363636...$

$10x = 1.363636...$

Subtract: $990x = 135$

$x = \frac{135}{990} = \frac{3}{22}$

1 mark
Align the recurring tails of 0.1363636...

Let $y = 0.2222...$

$10y = 2.2222...$

$y = 0.2222...$

Subtract: $9y = 2$

$y = \frac{2}{9}$

1 mark
Align the recurring tails of 0.2222...

$xy = \frac{3}{22} \times \frac{2}{9} = \frac{6}{198} = \frac{1}{33}$

1 mark
Exact fraction product completes the proof

(Total for Question 16 is 3 marks)

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